

STREAMLIT - POWERED DATA SCIENCE VISUALIZING CRYPTOCURRENCY MARKETS

A Project Report

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In partial fulfillment of the requirements for the award of the Degree of

BACHELOR OF TECHNOLOGY
In
COMPUTER SCIENCE AND ENGINEERING

By

**K.KOTESWARARAO
(21481A0598)**

**K.GANESH
(21481A0599)**

**J.KEERTHI LAHARI
(21481A0584)**

**K.VENKATA KALYANI
(21481A05A5)**

**G.THANMAI TEJA
(21481A0570)**

Under the guidance of
Mr.U.Vinod Kumar M.Tech,
Assistant Professor of CSE Department



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SESHADRI RAO GUDLAVALLERU ENGINEERING COLLEGE
(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)
SESHADRI RAO KNOWLEDGE VILLAGE
GUDLAVALLERU – 521356
ANDHRA PRADESH
2024-2025

SESHADRI RAO GUDLAVALLERU ENGINEERING COLLEGE

(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)

SESHADRI RAO KNOWLEDGE VILLAGE, GUDLAVALLERU

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

This is to certify that the project report entitled “**STREAMLIT - POWERED DATA SCIENCE VISUALIZING CRYPTOCURRENCY MARKETS**” is a bonafide record of work carried out by **K.KOTESWARARAO (21481A0598),K.GANESH(21481A0599) J.KEERTHILAHARI(21481A0584),K.VENKATAKALYANI(21481A05A5), G.THANMAITEJA(21481A0570)** under the guidance and supervision in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering of Jawaharlal Nehru Technological University Kakinada, Kakinada during the academic year 2024-25.

Project Guide
(Mr. U . Vinod Kumar)

Head of the Department
(Dr. M. BABU RAO)

External Examiner

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Team members

K.Koteswararao	(21481A0598)
K. Ganesh	(21481A0599)
J.Keerthi Lahari	(21481A0584)
K.VenkataKalyani	(21481A05A5)
G.Thanmai Teja	(21481A0570)

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ABSTRACT

The Streamlit -powered Data Science Visualizing Cryptocurrency Markets project is a user-centric web application designed to provide deep insights into cryptocurrency trends. By leveraging technologies such as Python, Streamlit, Matplotlib, Seaborn, and Base64, this platform enables users to interactively analyze cryptocurrency data with real-time visualizations. Hosted on Google Cloud Shell, the application ensures scalability, reliability, and accessibility for a diverse range of users, including crypto enthusiasts and investors.

The primary objective of this project is to offer a comprehensive and intuitive gateway to cryptocurrency analysis, bridging the gap between raw data and actionable insights. Users can explore price trends, market capitalization, trading volumes, and other key metrics through interactive charts and visualizations. Whether for educational purposes, investment strategies, or general awareness, this platform serves as an indispensable resource in navigating the dynamic cryptocurrency landscape.

Streamlit-powered data science visualizing cryptocurrency markets is a pioneering and user-centric platform meticulously crafted to immerse users in the captivating realm of cryptocurrencies. By harnessing state-of-the-art technologies such as Python, Streamlit, Matplotlib, Seaborn, and Base64, this application introduces an engaging and immersive experience for both cryptocurrency enthusiasts and astute investors. Hosted on the robust Google Cloud Shell infrastructure, it not only assures scalability but also reinforces reliability, offering users unfettered access to invaluable insights and interactive tools tailored for traversing the dynamic and ever- evolving cryptocurrency market. The fundamental goal of this project is to furnish an expansive and user- friendly gateway to the multifaceted world of cryptocurrency analysis. The Streamlit-powered data science visualizing cryptocurrency markets empowers users with a rich tapestry of interactive data visualizations and in-depth insights, thereby arming them with the essential instruments required to delve into and comprehend the complex cryptocurrency landscape. Whether you are an ardent cryptocurrency aficionado yearning to stay apprised of the latest trends or an astute investor in pursuit of data-driven decision-making, our web application stands as an invaluable and irreplaceable resource in the vast expanse of the cryptocurrency space.

Keywords - Crypto Visualization, Data Analysis, Streamlit, Cryptocurrency Insights,
MarketTrend

CHAPTER – 1

1.1 Introduction

Cryptocurrencies have revolutionized the global financial system, offering decentralized and secure alternatives to traditional banking. Since the inception of Bitcoin in 2009, thousands of digital currencies have emerged, creating a vast and complex market. Understanding this market requires advanced data analysis techniques and real-time monitoring tools. The Streamlit-powered Data Science Visualizing Cryptocurrency Markets project aims to simplify cryptocurrency analysis by providing an interactive and user- friendly platform for exploring real-time and historical market trends.

This project leverages Python, Streamlit, Matplotlib, Seaborn, and Base64 to offer an engaging experience for users. The application enables individuals to analyze crucial cryptocurrency metrics such as price trends, trading volumes, and market capitalization through visually appealing charts and graphs. Additionally, it provides insights into historical price movements and helps users identify patterns that can assist in investment decisions. By integrating modern web technologies, this project bridges the gap between raw data and meaningful insights, making cryptocurrency analysis accessible to both beginners and experienced investors.

With the rapid adoption of cryptocurrencies, there has been a growing demand for data-driven decision-making tools. Many investors and enthusiasts struggle to interpret complex market data, making it difficult to predict trends and make informed financial decisions. Existing tools often require advanced technical knowledge or are restricted to paid users. Our motivation behind this project is to create a free, open-source, and accessible platform that simplifies cryptocurrency analysis and provides real-time data visualization without requiring users to have a deep understanding of programming or financial analytics.

By using Google Cloud Shell, the platform ensures scalability and reliability, allowing users to access it from anywhere without the need for complex installations or configurations. Whether a user is a trader looking for short-term trends or an investor analyzing long-term market behavior, this tool provides valuable insights to support decision-making.

Our project, "Streamlit-powered data science visualizing cryptocurrency markets," is dedicated to providing users with a sophisticated and user-friendly platform to explore the world of cryptocurrencies. Through the innovative application of Python, Streamlit, Matplotlib, Seaborn, and Base64, we offer an immersive experience for crypto enthusiasts and investors. Hosted on the robust Google Cloud Shell infrastructure, our project guarantees scalability and reliability, ensuring users have access to valuable insights and interactive tools for navigating the dynamic cryptocurrency market. We are committed to demystifying the complexities of cryptocurrency analysis, making it accessible to individuals from all walks of life. Our mission is rooted in empowering users with the necessary tools and knowledge to understand and thrive in the ever- changing cryptocurrency landscape. Whether you are an enthusiast looking to stay informed or an investor seeking data-driven decision- making, our web application serves as your indispensable resource in the world of crypto. Our project stemmed from a shared passion for cryptocurrency and the realization that many individuals face challenges in comprehending and navigating the crypto market. The project's founders, driven by a deep fascination for digital assets and a desire to promote informed decision making, embarked on this journey to bridge the gap between the complexities of cryptocurrency and everyday users. The problem at hand revolves around the complexity and inaccessibility of cryptocurrency analysis for the average individual. Challenges include understanding market trends, interpreting data, and making informed decisions. Many potential investors are discouraged by the lack of user friendly tools and educational resources. The primary objective of this project is to provide a comprehensive and accessible gateway to the realm of cryptocurrencies analysis. The Streamlit-powered data science visualizing cryptocurrency markets empowers users with a range of interactive data visualizations and insights, equipping them with the necessary tools to explore and understand the ever-changing cryptocurrency landscape. Whether you are a crypto enthusiast looking to stay informed or an investor seeking data driven decision-making, our web application serves as your indispensable resource in the crypto space.

1.2 Objectives of the Project

The primary objective of this project is to design and develop a robust, user-friendly, and interactive web application for visualizing and analyzing cryptocurrency data. In recent years, cryptocurrencies have become a focal point in the financial world, with their high volatility, decentralized nature, and growing adoption. This project seeks to bridge the gap between complex crypto market data and user-friendly, interpretable insights through a streamlined interface. The following detailed goals guide the development and purpose of the project:

1. Development of an Interactive Web-Based Visualization Tool

At the core of the project is the creation of an interactive web-based platform that allows users to visualize cryptocurrency data effectively. This includes the integration of dynamic and responsive charts, bar plots, pie charts, and tables to represent various metrics such as price changes, market dominance, and trading volume. The platform is designed with an intuitive interface that caters to users ranging from beginners to expert investors, offering seamless data interaction.

2. Provision of Real-Time and Historical Market Data

The project is engineered to support both real-time updates and access to historical data of selected cryptocurrencies. Using APIs and web scraping techniques, it fetches live data on prices, percent changes across different time frames (1 hour, 24 hours, 7 days), and other essential metrics. This dual capability empowers users to observe current market conditions while also analyzing past trends to make informed decisions.

3. Tracking of Key Metrics for Informed Decision-Making

Users can monitor vital indicators such as price fluctuations, market capitalization, percent change over time, and trading volume. These metrics are crucial for making data-driven investment decisions. For example, users can quickly identify which cryptocurrencies are performing well in the short term or which have gained or lost market share, helping them adapt strategies based on trend analysis.

4. Accessibility and Scalability via Google Cloud Shell Hosting

The application is hosted on Google Cloud Shell, which ensures that the tool is both accessible and scalable. This cloud-based deployment eliminates the need for local installations, allows for instant launching in a web browser, and facilitates smooth upgrades or future feature additions. Additionally, Google Cloud Shell ensures that the application benefits from the security, reliability, and global infrastructure of Google Cloud.

5. Promotion of Data-Driven Insights in an Easy-to-Understand Format

The project prioritizes simplicity and clarity in its visual outputs to make complex data easily understandable for users. Visualizations like bar plots of percent changes, market cap comparisons, and pie charts for market dominance help users grasp market dynamics at a glance. The goal is to encourage and enable data-driven decision-making without requiring users to have a deep technical background or prior data science experience.

6. Encouraging Education and Awareness Among Users

Beyond functional utility, the platform aims to serve as an educational tool by helping users learn about cryptocurrency markets. By interacting with real-world data and observing patterns through visual means, users can better understand the mechanics of digital asset trading, volatility, and market behavior over different timeframes.

7. Support for Customization and Filter-Based Exploration

The project allows users to select specific cryptocurrencies, customize the currency unit (e.g., USD), and choose the time frame for percentage change analysis. These customizable input options enhance the usability of the platform by giving users control over what data they want to see and how they want it presented.

1.2 Problem Statement:

The cryptocurrency market is highly volatile, and predicting its movements requires the analysis of large datasets. Some of the major challenges faced by users include:

Complexity of Market Data: Cryptocurrency data is often unstructured and difficult to interpret without specialized knowledge.

Lack of User-Friendly Analytical Tools: Most data analysis platforms require programming skills or expensive subscriptions.

Need for Real-Time Insights: Due to rapid price fluctuations, users require up-to- date data visualization to make timely decisions.

Scalability and Accessibility: Many existing tools require local installations, making Inaccessible to user who lack technical expertise.

1.3 Scope of Research:

The scope of this research focuses on developing a Cryptocurrency Market Visualization Web Application that enables real-time data extraction, analysis, and interactive visualization using Streamlit, Pandas, Matplotlib, Seaborn, Beautiful Soup, and Crypto CMD.

Key Areas of Research

1. Cryptocurrency Market Trends Understanding price fluctuations, trading volumes, and market capitalization. Examining factors influencing crypto prices, such as regulations, supply-demand, and investor sentiment.
2. Web Scraping for Real-Time Data Extraction Efficient data retrieval using Beautiful Soup from Coin Market Cap. Challenges in scraping (anti-bot mechanisms, dynamic content loading).
3. Data Processing & Management Cleaning and structuring raw data using Pandas. Handling missing values and inconsistent price formats.
4. Data Visualization Techniques Utilizing Matplotlib and Seaborn for price trends, bar charts, and time-series analysis. Comparative analysis of multiple cryptocurrencies in real time. Interactive Web Application with Streamlit Deploying an intuitive user interface for easy data exploration. Real-time filtering and selection of cryptocurrencies.
5. Deployment & Scalability Hosting the Streamlit app on Google Cloud Shell. Performance considerations for real-time visualization. Comparative Analysis with Existing Solutions Examining limitations of API-based platforms like Binance and Coin Gecko. Advantages of a custom-built visualization tool over third-party platforms.

6. Research Contribution a lightweight, real-time visualization tool for tracking cryptocurrency prices. A no-API approach using direct web scraping, making it accessible without API limitations. Cloud-based deployment, ensuring easy access without local installations.

CHAPTER – 2

2.1 LITERATURE REVIEW

Cryptocurrency market analysis is an extensively researched domain, with numerous platforms like Coin Market Cap, Coin Gecko, and Binance providing real-time cryptocurrency data. These platforms primarily rely on APIs to offer data analytics, which, despite being efficient, come with constraints such as rate limits, paid access, and dependency on third-party services. Our research addresses these constraints by utilizing web scraping techniques, allowing for unrestricted data retrieval.

Web scraping has been widely used in financial applications to collect stock market data, forex trends, and cryptocurrency prices. Studies indicate that BeautifulSoup and Selenium are among the most effective tools for scraping structured data from financial websites. However, scraping financial data presents challenges, including website protection mechanisms, CAPTCHA verification, and dynamically loaded content. Our implementation considers these factors, ensuring optimized and sustainable data retrieval without violating website terms.

Visualization is a key component of financial research, and several academic studies highlight the effectiveness of different techniques. Matplotlib and Seaborn are widely used to generate time-series plots, heatmaps, and comparative charts for cryptocurrency price movements. While some advanced frameworks like Plotly and D3.js offer interactive features, they often require additional dependencies and extensive front-end development skills. Streamlit, on the other hand, simplifies web-based data visualization without requiring front-end expertise, making it a suitable choice for this project.

Streamlit has gained popularity in recent years due to its ability to create interactive web applications with minimal coding effort. Research shows that Streamlit is particularly useful for real-time data visualization and machine learning model deployment. Unlike traditional web frameworks, Streamlit integrates seamlessly with Python's data science libraries, making it an excellent tool for financial analytics. Our research leverages Streamlit to create a web-based dashboard where users can analyze cryptocurrency data dynamically.

Cloud-based deployment is another critical area in web application development. Research suggests that platforms like Google Cloud, AWS, and Azure offer scalable solutions for hosting

Python-based applications. Google Cloud Shell, in particular, provides a lightweight environment for running Streamlit applications without requiring complex setup procedures. Our project benefits from cloud-based deployment by ensuring high availability, easy accessibility, and minimal maintenance efforts.

Existing research highlights that while cryptocurrency tracking platforms provide valuable insights, they come with limitations such as restricted access and lack of customization. Web scraping has been identified as a viable alternative for real-time data extraction, but it requires careful handling to avoid detection and ensure compliance. Data visualization techniques using Matplotlib and Seaborn are well established, and Streamlit emerges as an efficient framework for interactive web applications. Finally, cloud deployment enhances accessibility and scalability, making our approach a robust and user- friendly solution for cryptocurrency analysis.

CHAPTER – 3

PROPOSED METHOD

3.1 Data Acquisition and Preprocessing

The first phase of the project focuses on collecting real-time cryptocurrency data. Coin Market Cap is selected as the primary data source due to its extensive and reliable market coverage.

Tools & Techniques Used:

- **Web Scraping:** Implemented using the BeautifulSoup library.
- **Data Extracted Includes:**
 - Cryptocurrency Name
 - Market Capitalization
 - Price Fluctuations
 - Trading Volume
 - Historical Trends

Anti-Bot Strategies:

- User-Agent Rotation
- Request Throttling
- Dynamic Content Handling

Storage:

- Data is structured and stored in Pandas Data Frames for further processing.

3.1.1 Data Processing

To ensure high data quality and readiness for visualization, the following preprocessing steps are applied:

- **Handling Missing Values:**
 - Imputed using interpolation techniques or removed if necessary.
- **Data Formatting:**
 - Prices and volumes are standardized into numerical formats.

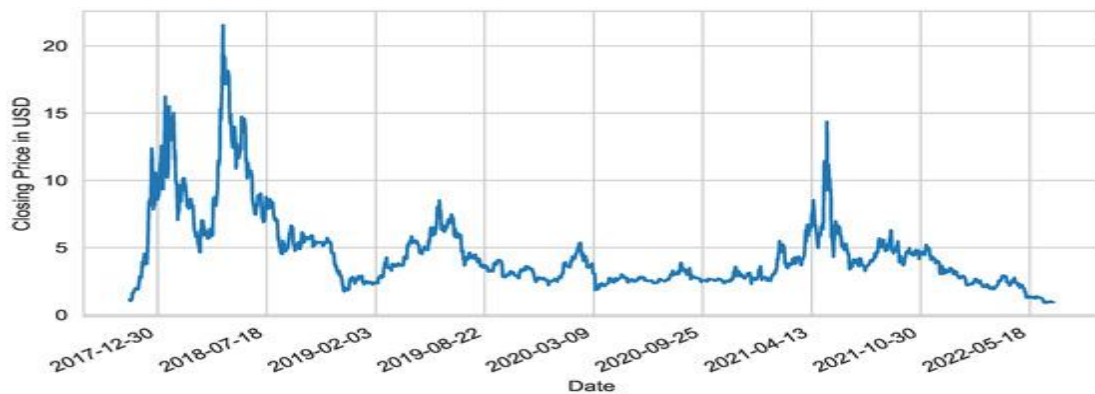
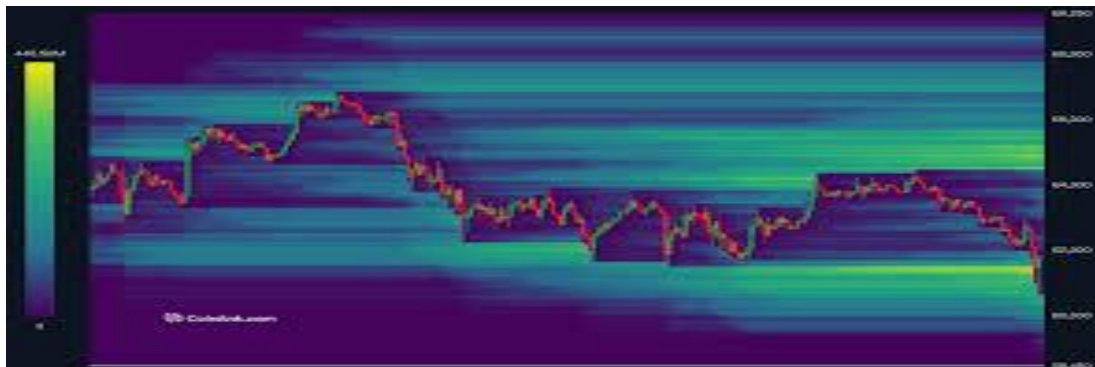
- **Outlier Detection:**
 - Erroneous extreme values are identified and filtered out.
- **Data Structuring:**
 - Cleaned data is indexed by date-time for time-series analysis using Pandas.

3.1.2 Data Visualization

Visual analytics are essential for interpreting complex market behavior. Matplotlib and Seaborn are used to build interactive and insightful visuals.

Types of Visualizations:

- **Time-Series Graphs:** Show daily price changes.
- **Bar Charts:** Compare market capitalization across cryptocurrencies.
- **Heatmaps:** Visualize correlations between market indicators.



3.2 Web Application Development

The cleaned and visualized data is integrated into a web application built with Streamlit.

Features of the Application:

- Simple, interactive user interface
- Dropdown filters for specific cryptocurrency selection
- Live updates reflecting real-time data changes

Advantages of Streamlit:

- No coding required for end users
- Dynamic exploration of data
- Fast, responsive, and intuitive UI

3.3 Deployment

The application is deployed on Google Cloud Shell, offering:

- **Scalability:** Efficient handling of large datasets
- **Cross-Device Access:** Accessible from any device with an internet connection
- **Minimal Setup:** No local installations required

Deployment Steps:

- Virtual environment setup
- Dependency installation
- Streamlit app execution via Google Cloud

3.4 Implementation Walkthrough

1. Dataset Collection

#	Name	Price	1h %	24h %	7d %	Market Cap	Volume(24h)	Circulating Supply	Last 7 Days
1	Bitcoin BTC	\$84,505.72	▼0.13%	▲1.72%	▼3.18%	\$1,677,041,039,401	\$28,597,951,100 338.46K BTC	19.84M BTC	
2	Ethereum ETH	\$1,876.98	▼0.06%	▲2.14%	▼8.54%	\$226,480,655,574	\$14,940,930,131 7.95M ETH	120.66M ETH	
3	Tether USDT	\$0.9999	▼0.00%	▲0.01%	▼0.03%	\$144,022,278,669	\$59,521,103,250 59.52B USDT	144.03B USDT	
4	XRP XRP	\$2.09	▼0.45%	▼0.31%	▼14.37%	\$121,986,932,490	\$4,062,189,840 1.94B XRP	58.2B XRP	
5	BNB BNB	\$600.59	▼0.26%	▼1.40%	▼4.59%	\$85,567,625,958	\$2,129,488,903 3.54M BNB	142.47M BNB	
6	Solana SOL	\$124.66	▲0.09%	▼1.11%	▼12.39%	\$63,890,810,897	\$2,986,437,131 23.98M SOL	512.5M SOL	
7	USDC USDC	\$0.9999	▼0.00%	▲0.00%	▼0.00%	\$60,077,023,599	\$9,161,817,353 9.16B USDC	60.07B USDC	
8	Dogecoin DOGE	\$0.1710	▼0.28%	▲1.25%	▼10.45%	\$25,432,915,744	\$1,165,335,105 6.80B DOGE	148.67B DOGE	

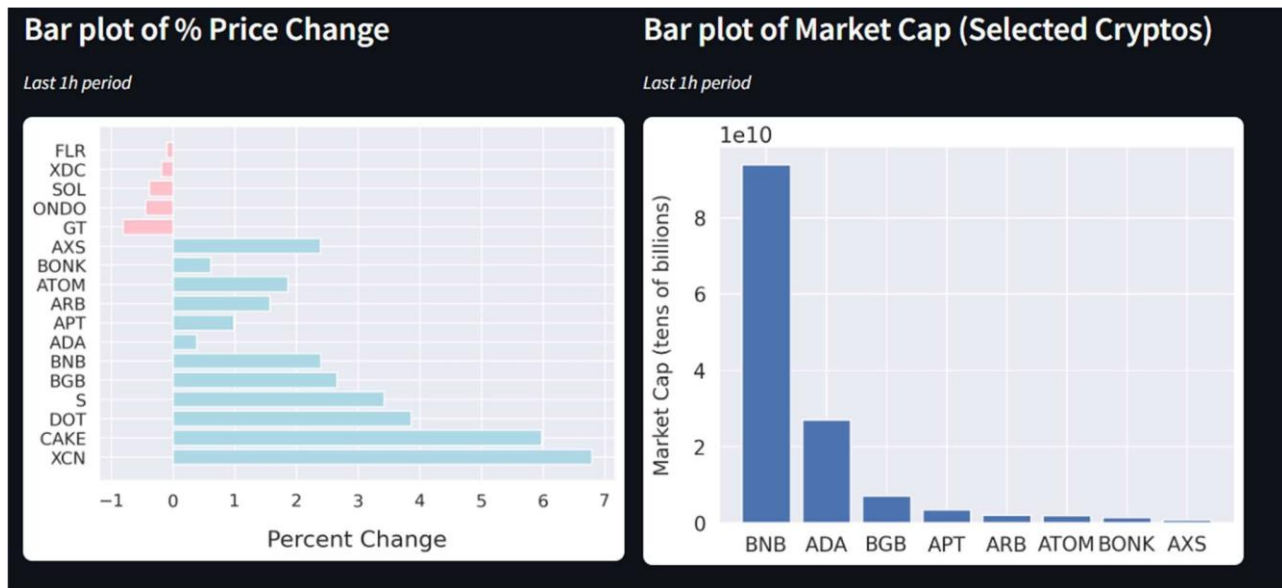
2. Code Implementation

```
cloudshell X + X
✓ Installation Succeeded
Installing matplotlib...
✓ Installation Succeeded
Installing seaborn...
✓ Installation Succeeded
Installing bs4...
✓ Installation Succeeded
Installing cryptocmd...
✓ Installation Succeeded
Installing dependencies from Pipfile.lock (179adf)...
All dependencies are now up-to-date!
Upgrading streamlit, pandas, matplotlib, seaborn, bs4, cryptocmd in dependencies.
Building requirements...
Resolving dependencies...
✓ Success!
Building requirements...
Resolving dependencies...
✓ Success!
Installing dependencies from Pipfile.lock (179adf)...
All dependencies are now up-to-date!
Installing dependencies from Pipfile.lock (179adf)...
(ProjectDS) keerthilahari2003@cloudshell:~/ProjectDS$ streamlit run crypto_price_app.py --server.port 8080 --browser.serverAddress https://8080-cs-507449746201-default.cs-asia-southeast1-ajrg.cloudshell.dev/?authuser=0&redirectedPreviously=true
[1] 1709
(ProjectDS) keerthilahari2003@cloudshell:~/ProjectDS$
Collecting usage statistics. To deactivate, set browser.gatherUsageStats to false.

You can now view your Streamlit app in your browser.

URL: http://https://8080-cs-507449746201-default.cs-asia-southeast1-ajrg.cloudshell.dev/?authuser=0:8080
```

3. Visualization Output



4 . Final Web Application Interface



3.5 Implementation Details

3.5.1 Model Training and Evaluation

This project does not involve machine learning model training. The emphasis is on real-time data processing, transformation, and insightful visualization.

Process Flow:

- **Data Collection:**
 - API calls to CoinMarketCap using the requests library
- **Data Cleaning & Processing:**
 - JSON responses are converted to structured Pandas DataFrames
 - Irrelevant fields are removed
 - Numerical data is formatted
- **Feature Selection:**
 - Selected features: price, market_cap, 24h_volume, percent_change over 1h, 24h, and 7d

3.5.2 Visualization & Evaluation

Using matplotlib, seaborn, and streamlit, the following visual representations are provided:

- **Bar Charts:** Price change across timeframes
- **Market Cap Charts:** Comparison across cryptocurrencies
- **Pie Charts:** Market share of Bitcoin, Ethereum, and Altcoins
- **Line Charts:** Historical trends retrieved using the crypto cmd library.

CHAPTER – 4

RESULTS AND DISCUSSION

4.1 Results



Figure-4.1 Code implement



Figure-4.2 Installations / execution



Figure-4.3

The form is titled "Input Options" and is set against a dark background. It contains three main sections: "Select currency for price" with a dropdown menu showing "USD"; "Cryptocurrency" with a row of red buttons for ADA, ARB, APT, AXS, and ATOM, and a second row for BGB, BONK, and BNB, followed by a coin icon and a dropdown arrow; and "Percent change time frame" with a dropdown menu showing "1h".

Figure-4.4

About

- Made By: <Your Name>
- Data source: [CoinMarketCap](#).
- Credit: Web scraper adapted from the Medium article [Web Scraping Crypto Prices With Python](#) written by [Bryan Feng](#).

Figure-4.5

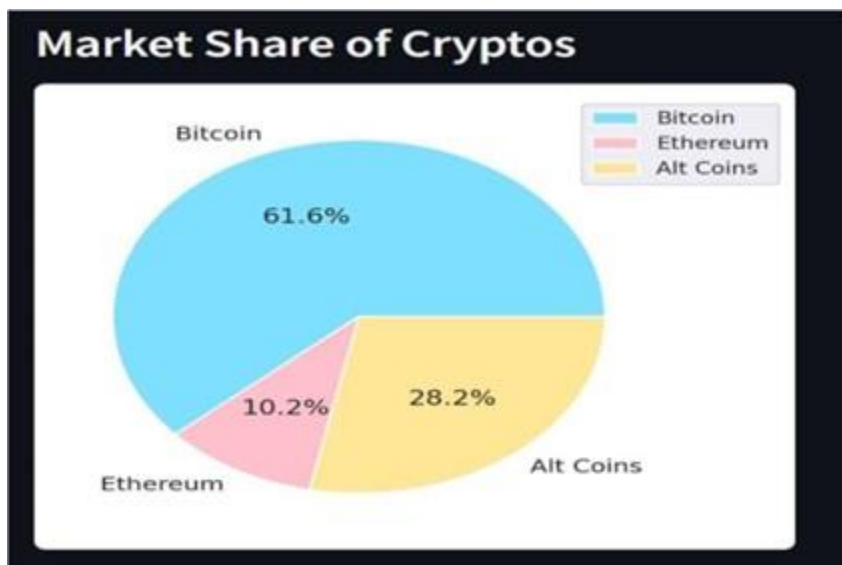


Figure-4.6



Figure-4.7

Tables

Price Data of Selected Cryptocurrencies

	coin_name	coin_symbol	market_cap	price	volume_24h
4	BNB	BNB	93,921,525,023.3987	59.194	2,305,525,163.7
8	Cardano	ADA	26,961,620,292.6173	0.766	1,431,040,827.6
21	Bitget Token	BGB	7,076,746,344.0091	5.8973	381,451,867.9
34	Aptos	APT	3,377,306,707.1329	5.8725	224,431,677.4
48	Arbitrum	ARB	2,013,341,053.1918	0.4635	218,403,576.4
50	Cosmos	ATOM	1,843,937,662.8181	4.7167	146,115,418.0
64	Bonk	BONK	1,310,541,791.728	0.00002	134,483,799.7
94	Axie Infinity	AXS	694,200,506.9722	4.3782	38,851,456.2

Figure-4.8



Figure-4.9

The analysis of cryptocurrency performance focuses on percent price change, market capitalization, and overall market share. The selected dataset captures the short-term price variations of several cryptocurrencies, providing insights into their volatility and investment potential. The visualizations presented in the study highlight the trends observed over a one-hour period, revealing the fluctuating nature of digital assets.

Percent Price Change Analysis:

The bar plot displaying the percentage price change over the last hour provides a comparative view of different cryptocurrencies' short-term volatility. Among the selected assets, BNB (Binance Coin) showed the highest positive change at approximately 2.4%, followed by Bitget Token (BGB) with a 2.6% increase. Conversely, some assets, such as Cardano (ADA) and Bitget Token (BGB), showed negative 24-hour changes, indicating a potential market correction or external influencing factors. This analysis highlights the rapid fluctuations in cryptocurrency values, which could be attributed to factors such as market demand, trading volume, and investor sentiment.

Additionally, certain cryptocurrencies, including Bonk (BONK) and Arbitrum (ARB), demonstrated moderate growth, suggesting stable but less aggressive price movements. The presence of both positive and negative changes within a short time frame reflects the unpredictable nature of the crypto market. Such trends emphasize the importance of real-time monitoring for traders and investors who engage in short-term trading strategies.

Market Capitalization Insights:

Market capitalization is a crucial metric that determines the overall value and stability of a cryptocurrency. The bar plot showcasing market cap for selected cryptocurrencies clearly indicates that BNB holds the highest market cap among the analyzed assets, exceeding \$80 billion. This signifies its strong position in the market and widespread investor confidence.

On the other hand, ADA (Cardano) follows with a significantly lower market cap, but still notable compared to lesser-known tokens like BGB, APT, and ARB. The disparity in market capitalization highlights the dominance of a few established cryptocurrencies over emerging altcoins. Assets with a higher market cap tend to have greater stability and resilience against price manipulation, making them preferable for long-term investment strategies.

The presence of assets like Bonk and Axie Infinity (AXS) with relatively low market capitalization suggests their speculative nature, where price movements can be heavily influenced by social media trends and community-driven trading. This supports the notion that market capitalization directly correlates with a cryptocurrency's risk level and potential for sudden price swings.

Market Share of Cryptocurrencies:

The pie chart illustrating market share distribution among Bitcoin, Ethereum, and Altcoins reveals the continued dominance of Bitcoin in the cryptocurrency landscape. With a market share of 61.6%, Bitcoin remains the most influential asset, dictating overall market trends and investor sentiment.

Ethereum follows with a 10.2% market share, reinforcing its role as the second-largest cryptocurrency, often regarded as the leading platform for decentralized applications (DApps) and smart contracts. Altcoins collectively hold 28.2% of the market, suggesting a growing interest in alternative digital assets, particularly in niche sectors like gaming, decentralized finance (DeFi), and Layer-2 scaling solutions.

The substantial market share held by Bitcoin and Ethereum suggests that they are the primary drivers of market movements. Historically, the price trends of major cryptocurrencies tend to impact the broader market, leading to synchronized price fluctuations across different assets. This phenomenon is often observed during periods of strong bullish or bearish trends, where investor sentiment shifts in response to Bitcoin's performance.

Implications for Investors and Traders:

The insights derived from the data emphasize the volatile nature of cryptocurrency investments. Short-term traders may find opportunities in assets with high price fluctuations, such as BNB and BGB, which demonstrated strong hourly gains. However, investors seeking long-term stability might prefer assets with higher market capitalization, such as BNB and ADA, as they are less prone to extreme price swings.

Additionally, the data suggests that altcoins with lower market capitalization, such as BONK and AXS, pose higher risks due to their susceptibility to price manipulation and speculative trading. While these assets can yield significant short-term gains, they also come with increased uncertainty.

The dominance of Bitcoin in market share further highlights its role as the "safe haven" of the crypto market. During periods of uncertainty, investors often shift capital toward Bitcoin and Ethereum, reducing exposure to riskier altcoins. This trend was evident in past market cycles, where Bitcoin's performance dictated the broader direction of the cryptocurrency market.

CHAPTER – 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

The integration of Python and Streamlit provides a powerful toolkit for crypto analysis and visualization. Leveraging Python's libraries such as Pandas, NumPy, and Matplotlib enables users to conduct in-depth data analysis, extracting valuable insights from cryptocurrency data. Streamlit's user-friendly interface further enhances the experience by facilitating dynamic dashboard creation and real-time data exploration. This allows users to customize workflows for various purposes, including monitoring price movements, analyzing trading volumes, and conducting sentiment analysis. By offering accessible and interpretable crypto market insights, this integration empowers users to make informed decisions in the fast-paced and dynamic world of cryptocurrency trading. Overall, the Python and Streamlit integration serves as a valuable resource for both seasoned traders and newcomers, providing the tools necessary to navigate the complexities of the crypto market effectively. Moreover, hosting the application on Google Cloud Shell adds an additional layer of accessibility and scalability. It eliminates platform dependency and allows for global usage without requiring local installation, making the tool practical and efficient for a wide range of users.

5.2 FUTURE SCOPE

Crypto Analysis and Visualization platform is promising, with several avenues for expansion and enhancement. As the cryptocurrency market continues to evolve, the platform can incorporate advanced analytical features such as sentiment analysis, machine learning-based price forecasting, and pattern recognition algorithms to provide users with deeper insights into market trends and dynamics. Additionally, the platform can explore integration with decentralized finance (DeFi) protocols and non-fungible token (NFT) markets to offer a comprehensive view of the broader cryptocurrency ecosystem. Enhanced collaboration and social networking features can foster a vibrant community of users, enabling knowledge sharing and idea generation. Moreover, expanding the platform's accessibility through mobile applications and multi-language support can broaden its user base globally. Continuous updating and adapting to emerging technologies and market trends will ensure that the Crypto Analysis and Visualization platform remains at the forefront of cryptocurrency analysis, serving as an indispensable tool for both enthusiasts and investors navigating the everchanging landscape of digital assets.

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SESHADRI RAO GUDLAVALLERU ENGINEERING COLLEGE

(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)
Seshadri Rao Knowledge Village, Gudlavalleru

Department of Computer Science and Engineering

Program Outcomes (POs)

Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problem
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions, component, or software to meet the desired needs.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader

Streamlit- Powered Data Science Visualizing
Cryptocurrency Markets
in diverse teams, and in multidisciplinary settings.

10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

PSO 1: Design, develop, test and maintain reliable software systems and intelligent systems. PSO

2 : Design and develop web sites, web apps and mobile apps.

PROJECT PROFORMA

Classification of Project	Application	Product	Research	Review
	√			

Note: Tick Appropriate category

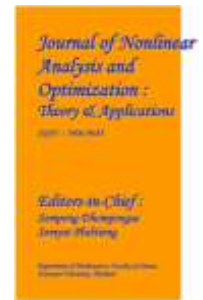
Project Outcomes	
Course Outcome (CO1)	Identify and analyze the problem statement using prior technical knowledge in the domain of interest.
Course Outcome (CO2)	Design and develop engineering solutions to complex problems by employing systematic approach.
Course Outcome (CO3)	Examine ethical, environmental, legal and security issues during project implementation.
Course Outcome (CO4)	Prepare and present technical reports by utilizing different visualization tools and evaluation metrics.

Mapping Table

CS3518: MAIN PROJECT															
Course Outcomes	Program Outcomes and Program Specific Outcome														
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12		PSO 1	PSO 2
CO1	3	3	1					2	2	2				1	1
CO2	3	3	3	3	3			2	2	2		1		3	3
CO3	2	2	3	2	2	3	3	3	2	2	2			3	
CO4	2		1		3				3	3	2	2		2	2

Note: Map each project outcomes with Pos and PSOs with either 1 or 2 or 3 based on level of mapping as follows:

1-Slightly (Low) mapped 2-Moderately (Medium) mapped 3-Substantially (High) mapped



STREAMLIT - POWERED DATA SCIENCE VISUALIZING CRYPTOCURRENCY MARKETS

U. Vinod Kumar, Assistant Professor, Seshadri Rao Gudlavalleru Engineering College (An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada), Seshadri Rao Knowledge Village, Gudlavalleru-521356, Andhra Pradesh, India

K. Koteswararao, J. Keerthi Lahari, G. Thanmai Teja, K. Venkata Kalyani, K. Ganesh, IV-B. Tech CSE Students Department of Computer Science and Engineering, Seshadri Rao Gudlavalleru Engineering College (An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada), Seshadri Rao Knowledge Village, Gudlavalleru-521356, Andhra Pradesh, India

Abstract—*Streamlit-powered data science visualizing cryptocurrency markets is a pioneering and user-centric platform meticulously crafted to immerse users in the captivating realm of cryptocurrencies. By harnessing state-of-the-art technologies such as Python, Streamlit, Matplotlib, Seaborn, and Base64, this application introduces an engaging and immersive experience for both cryptocurrency enthusiasts and astute investors. Hosted on the robust Google Cloud Shell infrastructure, it not only assures scalability but also reinforces reliability, offering users unfettered access to invaluable insights and interactive tools tailored for traversing the dynamic and ever- evolving cryptocurrency market. The fundamental goal of this project is to furnish an expansive and user- friendly gateway to the multifaceted world of cryptocurrency analysis. The Streamlit-powered data science visualizing cryptocurrency markets empowers users with a rich tapestry of interactive data visualizations and in-depth insights, thereby arming them with the essential instruments required to delve into and comprehend the complex cryptocurrency landscape. Whether you are an ardent cryptocurrency aficionado yearning to stay apprised of the latest trends or an astute investor in pursuit of data-driven decision-making, our web application stands as an invaluable and irreplaceable resource in the vast expanse of the cryptocurrency space.*

Index Terms– *Crypto Visualization (CV), Crypto Analysis(CA), Data Visualization(DV), StreamLit.*

I. INTRODUCTION

Our project, "Streamlit-powered data science visualizing cryptocurrency markets," is dedicated to providing users with a sophisticated and user-friendly platform to explore the world of cryptocurrencies. Through the innovative application of Python, Streamlit, Matplotlib, Seaborn, and Base64, we offer an immersive experience for crypto enthusiasts and investors. Hosted on the robust Google Cloud Shell infrastructure, our project guarantees scalability and reliability, ensuring users have access to valuable insights and interactive tools for navigating the dynamic cryptocurrency market.

We are committed to demystifying the complexities of cryptocurrency analysis, making it accessible to individuals from all walks of life. Our mission is rooted in empowering users with the necessary tools and knowledge to understand and thrive in the ever- changing cryptocurrency landscape. Whether you are an enthusiast looking to stay informed or an investor seeking data-driven decision- making, our web application serves as your indispensable resource in the world of crypto. Our project stemmed from a shared passion for cryptocurrency and the realization that many individuals face challenges in comprehending and navigating the crypto market. The project's

founders, driven by a deep fascination for digital assets and a desire to promote informed decision-making, embarked on this journey to bridge the gap between the complexities of cryptocurrency and everyday users.

The problem at hand revolves around the complexity and inaccessibility of cryptocurrency analysis for the average individual. Challenges include understanding market trends, interpreting data, and making informed decisions. Many potential investors are discouraged by the lack of user-friendly tools and educational resources.

The primary objective of this project is to provide a comprehensive and accessible gateway to the realm of cryptocurrencies analysis. The Streamlit-powered data science visualizing cryptocurrency markets empowers users with a range of interactive data visualizations and insights, equipping them with the necessary tools to explore and understand the ever-changing cryptocurrency landscape. Whether you are a crypto enthusiast looking to stay informed or an investor seeking data-driven decision-making, our web application serves as your indispensable resource in the crypto space.

II. RELATED WORK

Streamlit-powered data science visualizing cryptocurrency markets is a multifaceted project that merges data exploration, cryptography, and interactive presentation techniques. At its core, this project entails dissecting various aspects of digital currencies such as Bitcoin and Ethereum. This exploration includes delving into price trends, trading volumes, market capitalization, and blockchain data, among other metrics. Python, with its comprehensive suite of libraries like Pandas, NumPy, and Matplotlib, offers a robust ecosystem for data manipulation and visualization, rendering it an ideal platform for this endeavor.

To commence the analysis, data acquisition and preprocessing are paramount. This involves sourcing data from APIs provided by cryptocurrency exchanges, blockchain explorers, or historical data repositories. Subsequently, the collected data undergoes preprocessing stages which entail cleaning, filtering, and formatting to render it amenable for analysis and visualization.

Sentiment analysis constitutes another integral aspect of cryptocurrency analysis. By scrutinizing social media posts, news articles, and forum discussions, analysts can gauge public sentiment towards various cryptocurrencies.

Machine learning techniques serve as potent allies in predicting cryptocurrency prices or detecting market trends. Regression models, time series analysis, and clustering algorithms can be leveraged to uncover patterns within cryptocurrency data. Python's scikit-learn and TensorFlow libraries equip developers with the necessary arsenal to construct and train machine learning models effectively. Streamlit emerges as a pivotal component for crafting interactive visualizations and dashboards to showcase cryptocurrency analysis results. With its intuitive interface and declarative syntax, Streamlit facilitates the rapid prototyping and deployment of data-driven applications. By harnessing Streamlit's capabilities, developers can create dynamic platforms that provide real-time updates and engage users with compelling visualizations.

Drawbacks in existing system:

Crypto analysis tools come with their own set of advantages and limitations. While these tools are often tailored for specific types of analysis, such as tracking transactions or identifying patterns, they may not offer a comprehensive view of the entire crypto landscape. Moreover, the accuracy of these tools hinges greatly on the quality of the data they receive. Inaccuracies or gaps in the data can lead to misleading analysis results, undermining their usefulness.

Privacy concerns also loom over some crypto analysis tools. Certain tools may compromise user privacy by collecting personal or transactional data without proper consent or security measures in place. Additionally, processing large volumes of data in real-time poses a challenge for some tools, potentially causing delays in analysis and overlooking critical insights.

Another barrier to entry is the complexity of these tools, which often require a certain level of expertise to interpret correctly. This complexity can deter novice users or those lacking a deep understanding of crypto from effectively utilizing these tools.

Furthermore, the regulatory landscape surrounding cryptocurrencies is rapidly evolving, and some tools may struggle to keep pace with these changes. This could expose users to compliance risks if the tools fail to adhere to the latest regulations and standards. In light of these considerations, while crypto analysis tools offer valuable insights, users must approach their use with caution and awareness of their limitations.

III. PROPOSED WORK

The project "Streamlit-powered data science visualizing cryptocurrency markets" aims to revolutionize cryptocurrency analysis and visualization by offering a cutting-edge solution to address existing challenges. Key features of the proposed system include comprehensive data analysis and visualization tools powered by Python, Streamlit, Matplotlib, Seaborn, and Base64. These tools empower users to delve into cryptocurrency market trends, facilitating informed decision-making.

Moreover, the project emphasizes a user-friendly interface, catering to both seasoned cryptocurrency enthusiasts and newcomers alike. The intuitive design enables users to explore and analyze data effortlessly. Ensuring data security and user privacy, the system prioritizes implementing secure practices for online transactions and data protection, fostering a safe environment for user engagement.

Furthermore, the project integrates educational resources, providing insights into cryptocurrency trends to enhance users' knowledge and understanding of the crypto market. Hosted on Google Cloud Shell, the system ensures scalability and accessibility, enabling users to access the platform from diverse devices and locations. Overall, "Streamlit-powered data science visualizing cryptocurrency markets" presents a holistic solution to empower users in navigating the complexities of the cryptocurrency landscape.

A. Process flow and Architecture

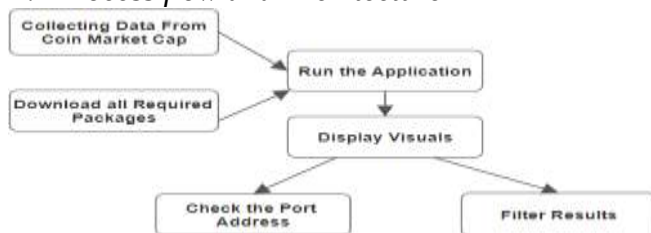


Fig. 1. Process Flow and Architecture.

The project begins with data collection from crypto APIs or databases, retrieving information on various cryptocurrencies. Next, the data undergoes preprocessing, including cleaning and the calculation of additional metrics such as moving averages or volatility. Analysis follows, where trends, correlations, and other insights are extracted from the data. With the data prepared, Streamlit is integrated to create an interactive web-based dashboard. This dashboard allows users to select specific cryptocurrencies, adjust date ranges, and explore different visualizations, such as price trends and volume charts. Through the Streamlit app, users can interact with the data, gaining valuable insights into the crypto market's trends and behaviors. The entire process aims to provide a user-friendly and informative tool for understanding and analyzing cryptocurrency data.

B. Dataset

#	Name	Price	1h %	24h %	7d %	Market Cap	Volume(24h)
1	Bitcoin BTC	\$64,278.18	+0.06%	+5.12%	+3.22%	\$1,284,982,475,860	\$81,404,736,106 301,630 BTC
2	Ethereum ETH	\$3,838.15	+0.37%	+7.84%	+10.30%	\$364,785,313,023	\$34,262,540,957 11,338,387 ETH
3	Tether USDT	\$1.00	+0.00%	+0.05%	+0.00%	\$187,770,756,257	\$134,128,370,987 134,042,170,243 USDT
4	BNB BNB	\$555.25	+0.74%	+7.99%	+5.52%	\$82,999,074,375	\$2,527,725,656 4,511,008 BNB

Fig. 2. Sample Dataset Images.

C. Evaluation Metrics

For a crypto analysis and visualization project, several evaluation metrics can be considered depending on the specific goals and focus of the analysis. Here are some common evaluation metrics:

Precision and Recall: Useful for classification models that predict buy/sell signals. Precision measures the accuracy of positive predictions, while recall measures the ability to find all positive instances.

A bar chart titled "Model Metrics" with a y-axis labeled "Score" ranging from 0.0 to 1.0. The x-axis lists four metrics: Precision, Recall, F1-score, and Accuracy. Each metric has a corresponding blue bar that reaches the 1.0 mark on the y-axis.

Metric	Score
Precision	1.0
Recall	1.0
F1-score	1.0
Accuracy	1.0

IV. RESULTS AND DISCUSSIONS

Figure 1 consists of two bar plots. The left plot, titled 'Bar plot of % Price Change', shows the percentage change in price for five cryptocurrencies: BTC, ETH, BNB, DASH, and ADA. The y-axis represents the percentage change, ranging from 0 to 100. The x-axis represents the cryptocurrency. The right plot, titled 'Bar plot of Market Cap (Selected Cryptos)', shows the market capitalization in USD for the same five cryptocurrencies. The y-axis represents the market cap in USD, ranging from 0.0 to 2.4.

Cryptocurrency	% Price Change	Market Cap (USD)
BTC	~85	~2.3
ETH	~45	~0.4
BNB	~15	~0.1
DASH	~10	~0.05
ADA	~10	~0.05

The screenshot shows a dashboard with two main sections. The left section, titled 'Input Options', contains a 'Market currency base price' dropdown set to 'USD', a 'Current currency' dropdown set to 'BTC', and a 'Market base price' dropdown set to 'USD'. The right section, titled 'Market Share of Cryptos', displays a pie chart showing the market share of four cryptocurrencies: Bitcoin (45.2%), Ethereum (23.7%), Bitcoin Cash (18.4%), and Litecoin (12.7%). The bottom right section, titled 'BTC over the last 30 days', shows a line chart of the BTC price over the last 30 days, with the price fluctuating between approximately 1000 and 1500 USD.

Fig. 6. Crypto Market Share.



Fig. 7. Downloadable CSV Data.

V. CONCLUSION

The integration of Python and Streamlit provides a powerful toolkit for crypto analysis and visualization. Leveraging Python's libraries such as Pandas, NumPy, and Matplotlib enables users to conduct in-depth data analysis, extracting valuable insights from cryptocurrency data.

Streamlit's user-friendly interface further enhances the experience by facilitating dynamic dashboard creation and real-time data exploration. This allows users to customize workflows for various purposes, including monitoring price movements, analyzing trading volumes, and conducting sentiment analysis.

By offering accessible and interpretable crypto market insights, this integration empowers users to make informed decisions in the fast-paced and dynamic world of cryptocurrency trading. Overall, the Python and Streamlit integration serves as a valuable resource for both seasoned traders and newcomers, providing the tools necessary to navigate the complexities of the crypto market effectively.

VI. FUTURE SCOPE

Crypto Analysis and Visualization platform is promising, with several avenues for expansion and enhancement. As the cryptocurrency market continues to evolve, the platform can incorporate advanced analytical features such as sentiment analysis, machine learning-based price forecasting, and pattern recognition algorithms to provide users with deeper insights into market trends and dynamics. Additionally, the platform can explore integration with decentralized finance (DeFi) protocols and non-fungible token (NFT) markets to offer a comprehensive view of the broader cryptocurrency ecosystem. Enhanced collaboration and social networking features can foster a vibrant community of users, enabling knowledge sharing and idea generation. Moreover, expanding the platform's accessibility through mobile applications and multi-language support can broaden its user base globally. Continuous updating and adapting to emerging technologies and market trends will ensure that the Crypto Analysis and Visualization platform remains at the forefront of cryptocurrency analysis, serving as an indispensable tool for both enthusiasts and investors navigating the ever-changing landscape of digital assets.

VII. ACKNOWLEDGEMENT

We Thank the Department of Computer Science and Engineering, GEC for supporting to carry out this work in all times.

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